

An AI Emulator of Holton-Mass Stochastic Sudden Stratospheric Warming with Faithful Dynamical Representation and an Interpretable Latent Structure

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1. Background

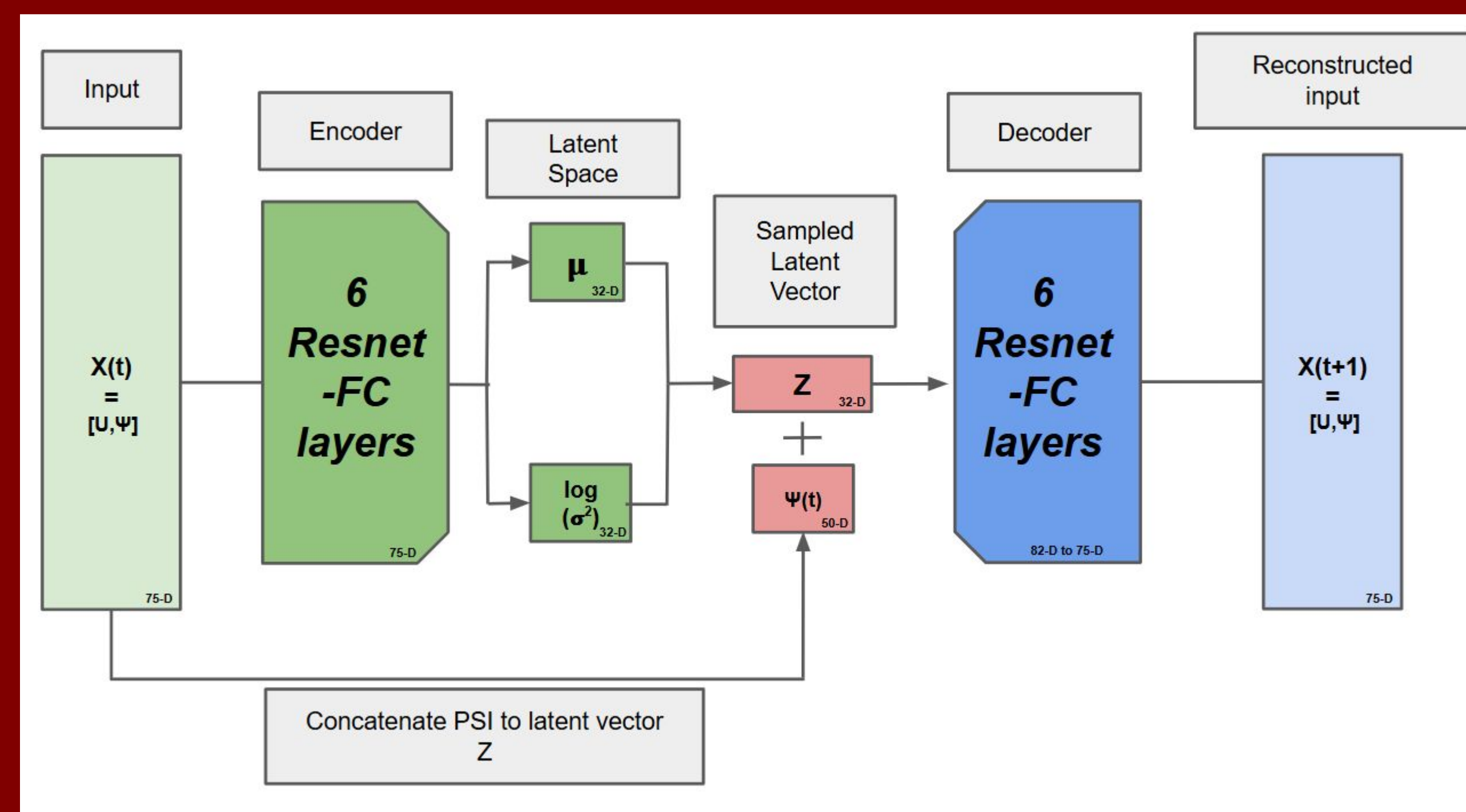
The polar vortex is an important component of the global climate system, influencing mid-latitude weather patterns. A notable phenomena in the northern stratosphere is a Sudden Stratospheric Warming (SSW). Sudden disruptions to the polar vortex are linked to outbreaks of extreme weather, including severe cold snaps and SSW's making its accurate prediction a key goal for forecasting.

2. Methods

2.1 Data

The data used for the emulator training is created by a 300k day simulation of the physical Holton-Mass Model. The length of the numerical time step is 0.005 of a day, which is then saved once per day. The physical dataset is comprised of 3 atmospheric variables split over 25 equal-sized levels of the stratosphere, and the height of the stratosphere is assumed to be 70km. The atmospheric variables are zonal wind (U), the real part of the stream function (Re(Ψ)), and the imaginary part of the stream function (Im(Ψ)). They are stacked such that the first 25 rows are U, the next 25 rows are Re(Ψ), and the last 25 rows are Im(Ψ). It follows that the dataset has a high number of degrees of freedom, $d=75$. Depending on the computer this data can take hours to generate.

2.2 Model Architecture



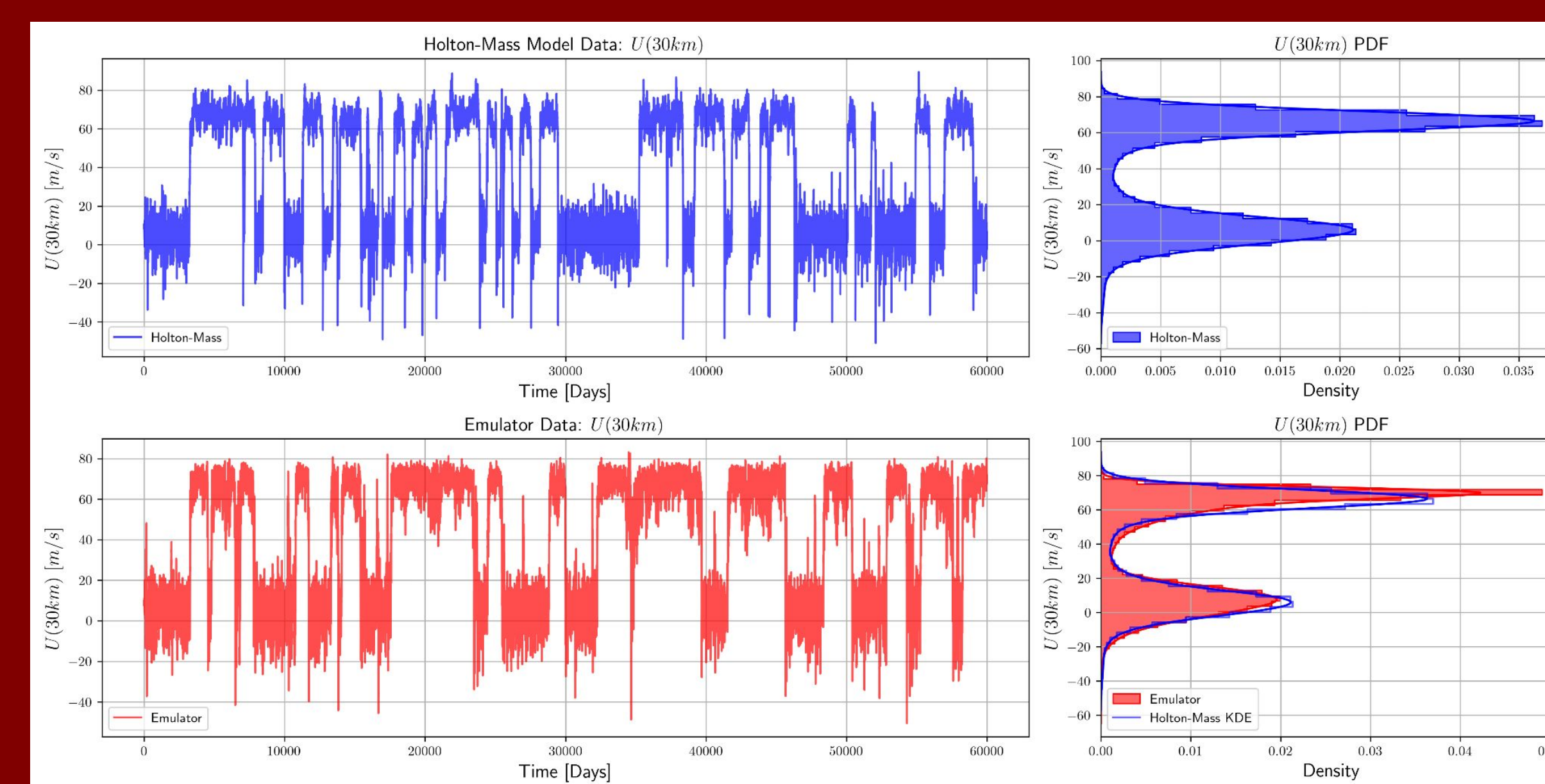
2.3 Model Training

- Loss Function:** Equally weighted Smooth L1 loss and KL loss
- Training:** Batch size of 1024 time steps, learning rate of 10e-4, 1500 epochs
- KL Annealing:** Provides a partial solution to posterior collapse e.g. when the latent space is negligible in inference
- Finetuning:** Due to the emulator favoring state A, we modified the physical dataset by removing time steps from the middle of state A. This improved the KL error between the two zonal wind PDFs of the physical and emulator models
- Choosing Best Model:** Computed the euclidean distance for the exponential fit of return periods, duration of return periods, and KL divergence error between PDFs e.g. how much information is lost as the emulator tries to approximate the physical PDF

3. Results

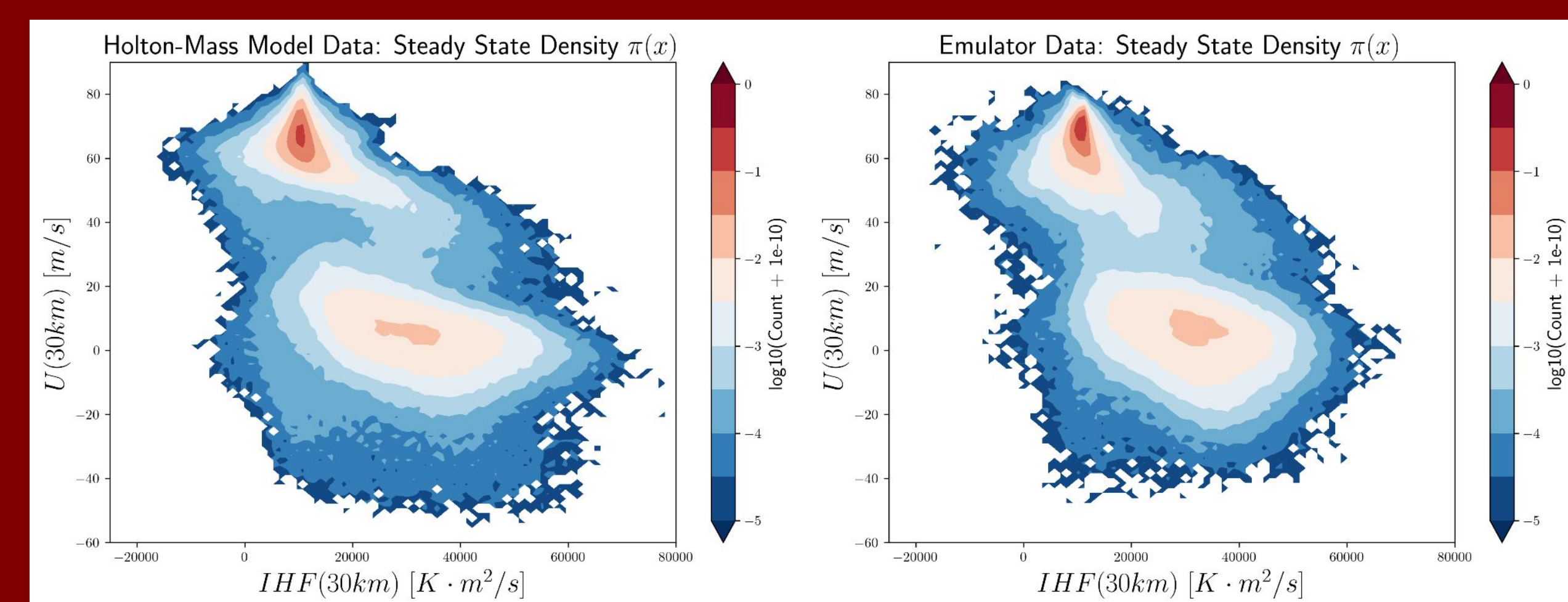
3.1 Long-Term Emulation of Vortex Dynamics

- The CVAE Emulator reproduces the full 300,000 day Holton-Mass time series of U(30km), faithfully capturing alternating strong and weak vortex regimes
- Transitions between states ($A \rightarrow B$ and $B \rightarrow A$) occur with realistic frequency and duration, matching the stochastic variability of the reference model



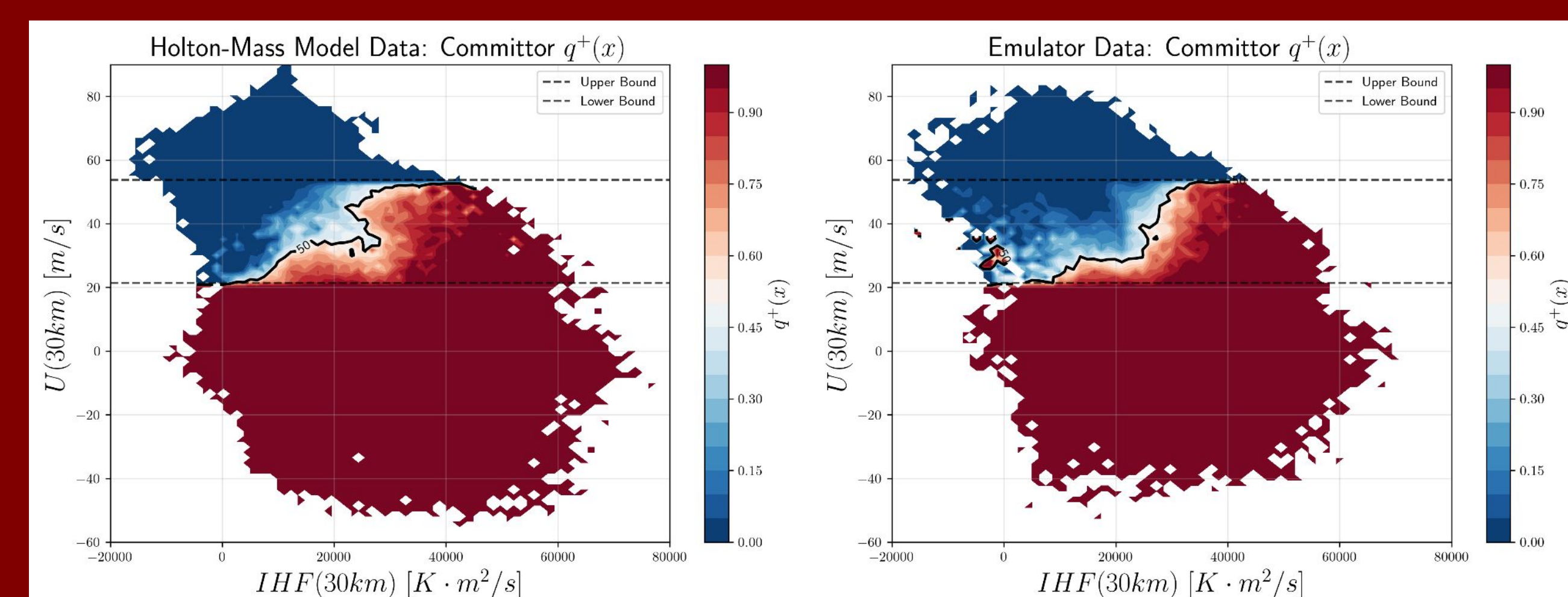
3.2 Steady-State Density and Regime Statistics

- Long free-runs of the emulator reproduce the Holton-Mass steady-state probability densities in one and two dimensions
- No false attractors or collapse into a single mode appear, demonstrating that the CVAE captures the stochastic equilibrium of the system



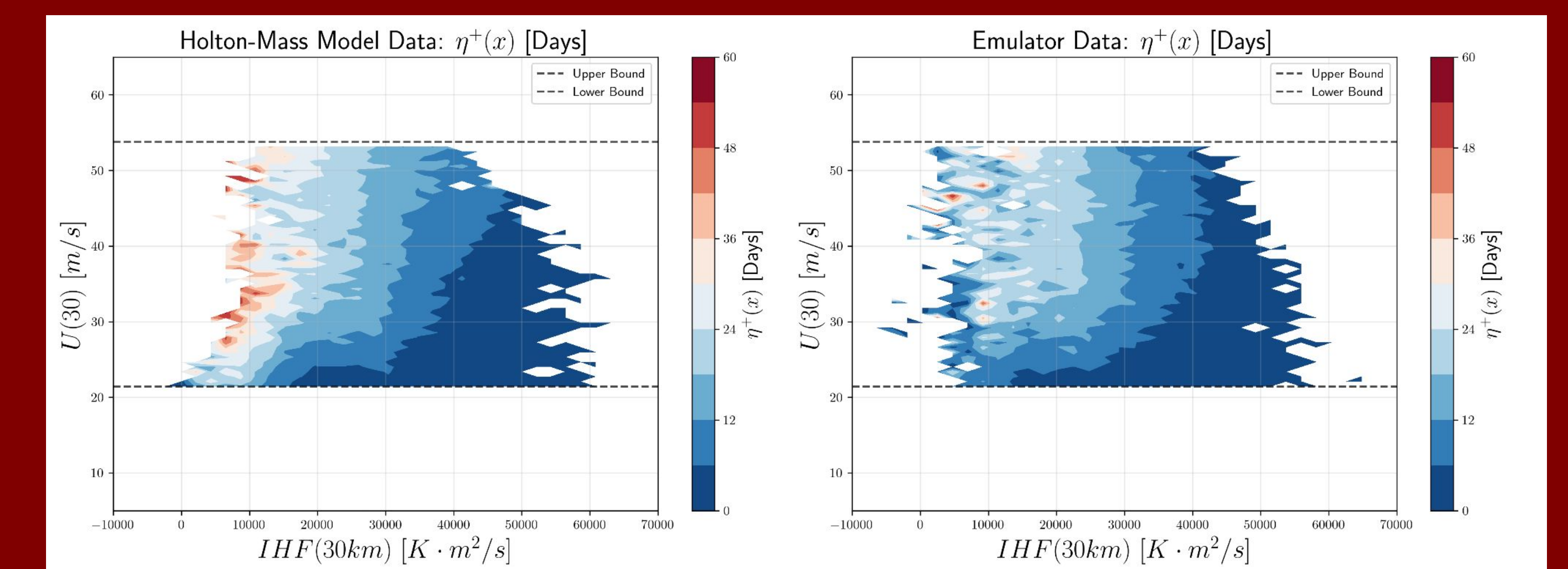
3.3 Committor Metrics of SSW

- The plot displays the probability of an SSW event happening with respect to zonal wind (U) and integrated heat flux from two long 300k days dataset of physical and emulator data, respectively
- In the transition state, which is delimited by the bounding dashed lines in the two plots, there is a clear relationship between the zonal wind and IHF, displaying that the model has learned that this is an important factor when deciding to transition or not



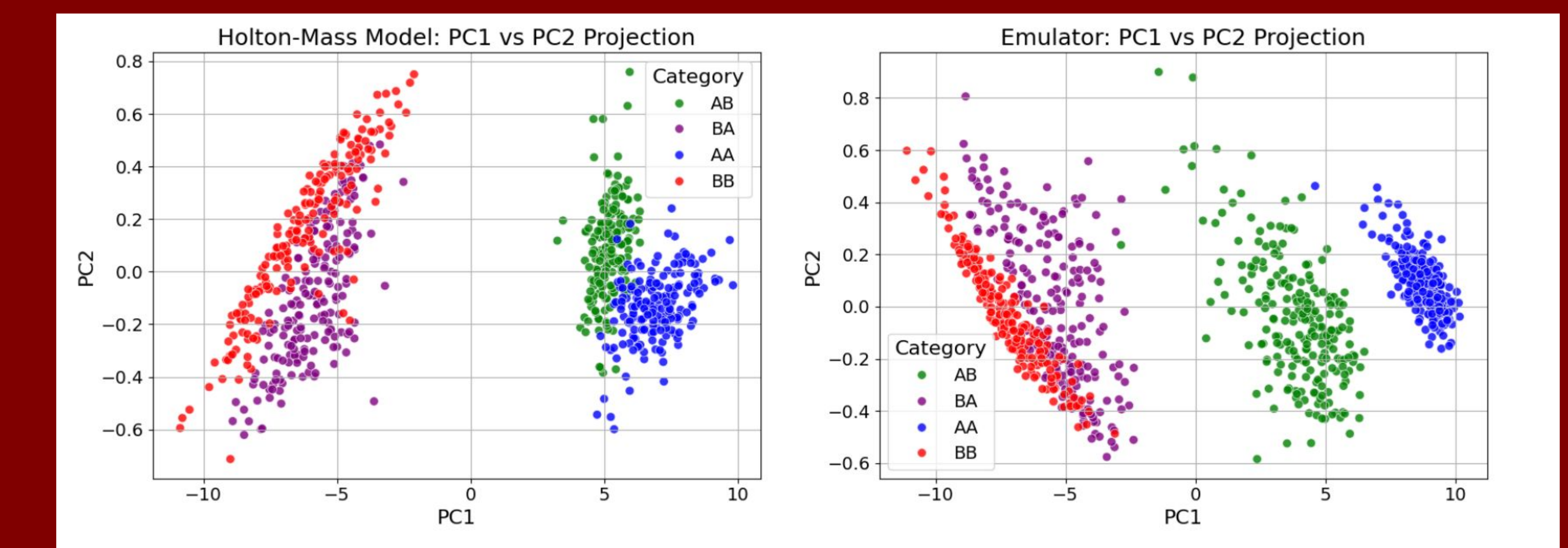
3.4 Lead Time Metrics of SSW

- The lead time of a transition is the number of days before a point within the transition region, $T = A^c \cap B^c$, transitions into state B
- As U(30) decreases and IHF increases, we see a positive correlation toward a lower lead time for transitions
- The emulator is able to capture the distribution of lead time for the physical model, including longer lead times



3.4 Interpretable Latent Structure (PCA)

- Principal-component analysis of latent means reveals four well-separated clusters that correspond to a strong/stable vortex, weak/stable vortex, pre-transition ($A \rightarrow B$), and Pre-transition ($B \rightarrow A$)
- Points near cluster boundaries coincide with imminent transitions in the time series, indicating the latent space encodes early-warning information
- This separation arises **without regime labels**, showing that the CVAE learns physically meaningful structure directly from the data



4. Acknowledgements

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